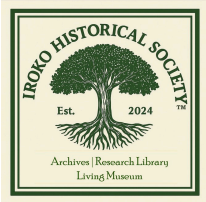
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IROKO HISTORICAL SOCIETY · WHITE PAPER · SECOND EDITION

The Iroko Framework

Semantic Vocabularies for Governing Access to Afro-Atlantic Sacred Knowledge Systems

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VOCABULARY
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Second edition · Describes framework release v1.4.0
Designed infrastructure and planned v2.0.0 changes are identified in the text.

About this edition. The first edition of this white paper was published in 2026 under DOI 10.5281/zenodo.18826673. This second edition describes the same framework release, v1.4.0, as it is actually published in the vocabulary files, and corrects the first edition where the two had drifted apart (Appendix D). It adds the framework's first two governed corpora, Ewé and the Hyatt Hoodoo corpus (Section VIII), the reference architecture for governed delivery (Section IX), and the finer access grain planned for v2.0.0 (Section XII). Throughout, three labels mark the status of what is described: PUBLISHED for what the vocabulary files contain today, DESIGNED for infrastructure that is specified but not yet built, and PLANNED, V2.0.0 for vocabulary changes under consideration.

SECTION I

Executive Summary

Afro-Atlantic sacred knowledge systems are among the most extensively documented and most inadequately described bodies of knowledge in the world's cultural heritage institutions. The materials exist. The infrastructure to steward them on the terms of the communities that hold them has lagged far behind.

The Iroko Historical Society is a postcustodial cultural heritage complex comprising a digital archive, a research library, and a living museum, built to serve Afro-Atlantic sacred communities as governing authorities over their own knowledge rather than as subjects of institutional description. The Society holds all three components under one governance logic without collapsing the distinctions between them. The archive governs restricted sacred materials under community authorization protocols. The research library provides structured scholarly access to public and community-authorized holdings. The living museum holds active sacred tradition at the center of its institutional identity.

The Iroko Framework is the semantic infrastructure that binds the Society together. Published at ontology.irokosociety.org under a CC0 dedication, release v1.4.0 comprises sixteen modules and two alignment modules, with 101 classes, 413 properties, 76 concept schemes, and 655 concepts.[1] The modules are organized across a Foundation layer, a Governance Layer of five composable sovereignty modules, and a Domain Layer of ten tradition-specific vocabularies. Six design principles govern the whole: field-level access control across six tiers, contested knowledge modeling through assertion reification, provenance tracking through oral transmission chains and fieldwork events, sovereign agency for sacred agents and community governance bodies, postcustodial design as a structural default, and semantic interoperability with established standards.

The framework's most consequential architectural decision is the separation of vocabulary publication from data publication. The vocabulary is public and available to any institution working with Afro-Atlantic sacred materials. The records described with it are governed:

disclosed field by field, to the people the holding community authorizes. Any institution can use the language. Only authorized parties can read what has been said in it.

This edition reports two governed corpora. Ewé, a fifty-record pilot of sacred plant knowledge drawn from Pierre Verger's ethnobotanical fieldwork, showed that access defaults published with each property can be evaluated against individual records in code, and why that evaluation must never depend on hand maintenance.[2] The Hyatt corpus, 20,279 records extracted from Harry Middleton Hyatt's five-volume *Hoodoo, Conjuration, Witchcraft, Rootwork*, is the first at scale and the first in which disclosure has to be decided below the level of the record: field by field and passage by passage within a single interview.[3] Hyatt is also the corpus that makes the remaining work concrete. The governed application it requires, a PostgreSQL canonical store, a Next.js interface with server-side enforcement, and filtered exports generated from approved fields only, is fully specified and is the subject of a twelve-month pilot reviewing 321 records (Section IX).

The institutions that need this infrastructure are identifiable and numerous: HBCUs with African diaspora collections, Caribbean national archives, museum repatriation programs, digital humanities centers, community-based temple archives, and theological seminaries with African and diaspora holdings. Support enables three phases of work: governed delivery and community authorization protocols, beginning with the Hyatt pilot; fieldwork, framework validation, and archivist training; and full operational capacity across the archive, library, and museum. The vocabulary is published. The design for governed delivery is complete. What remains is to build it, prove it on a bounded corpus, and open it to the institutions that need it.

SECTION II

The Institution: Why the Form Requires New Infrastructure

> *The preservation of Afro-Atlantic sacred knowledge systems has always been a problem of institutional form.*

Western cultural heritage institutions were built on assumptions of public access, neutral description, and institutional custody that sit uneasily with traditions governed by initiation, lineage authority, and community-held sovereignty over sacred knowledge. The result has been a century of Afro-Atlantic sacred materials held in institutions that struggle to describe them ethically, govern access to them appropriately, or return them meaningfully even when the will to do so exists. The materials survive. The infrastructure to steward them has not kept pace.

The Iroko Historical Society was founded to build that infrastructure. It operates as a postcustodial cultural heritage complex of three interconnected components. Each serves a distinct function and holds a distinct relationship to the knowledge in its care. Community-governed institutions already exist in many forms, from terreiro museums held by Candomblé houses in Bahia to Indigenous community archives built on platforms such as Mukurtu.[4][5] The Society's contribution is to hold

archive, library, and museum together under one governance logic designed by practitioners, across many Afro-Atlantic traditions at once, and to encode that logic in a shared vocabulary that other institutions can adopt.

DIGITAL ARCHIVE

Governs restricted sacred materials under a tiered access system calibrated to community authorization protocols.

`iroko:AccessPolicy` `iroko:StewardshipMandate` `iroko:DisclosureRestriction` `iroko:FieldworkEvent`

RESEARCH LIBRARY

Provides structured scholarly access to Afro-Atlantic sacred knowledge systems across traditions, geographies, and historical periods.

`iroko:ContestedAuthority` `iroko:Testimony` `iroko:Assertion` `iroko:Authority`

LIVING MUSEUM

Holds the living reality of active sacred traditions at the center of its institutional identity, as operational fact rather than curatorial stance.

`iroko:RitualEvent` `iroko:ManifestationEvent` `iroko:SacredInstrument` `iroko:RefusalEvent`

These three components create demands that no single existing standard addresses together. The archive requires access control fine enough to distinguish what may be known about a sacred material from who may know it, and under what conditions. The research library requires contested knowledge modeling able to document lineage disputes, variant traditions, and multiple authoritative perspectives without forcing resolution. The living museum requires sovereign agency infrastructure that models community authority, refusal, and stewardship mandates as institutional functions rather than policy exceptions.

The Iroko Framework exists because the institution required it. Every module, access tier, and governance mechanism in the vocabulary answers a specific descriptive demand the Society encountered in practice. The framework is not a technical contribution in search of an application. It is the infrastructure the institution needed in order to work.

SECTION III

The Problem Existing Standards Leave Open

The metadata standards that govern most cultural heritage institutions were built for a particular kind of institution: one that holds materials it owns, describes them in its own voice, and makes them available to credentialed researchers. The Dublin Core element set provides fifteen generic elements for describing a document, image, or artifact.[6] Encoded Archival Description structures finding aids. Resource Description and Access governs bibliographic description. These standards do what they were designed to do. Applied to Afro-Atlantic sacred materials, they leave four needs open, and the gaps are structural rather than incidental.

The Access Architecture Gap

Existing standards can record that access is restricted, and some can attach that statement at fine levels of description. DCMI Metadata Terms includes an `accessRights` property for a resource.[7] EAD's `<accessrestrict>` element may appear at every level of a finding aid, from the whole collection down to a single file or item.[8] What these provide is a statement, usually prose, about a unit of description. They do not provide a shared, tiered vocabulary that a system can act on property by property within one record. A sacred corpus needs exactly that. A single object may have a public name, a scholarly description available to credentialed researchers, a ritual context visible to initiates, and preparation knowledge restricted to elders with lineage authority. An interview in the Hyatt corpus may carry a public citation, a restricted informant reference, and passages of testimony that warrant different levels of disclosure. Machine-actionable policy languages such as the W3C's ODRL can express permissions and prohibitions over assets, but they are domain-neutral and carry no vocabulary for initiation, eldership, or lineage authority.[9] The Iroko Framework publishes six access tiers as a SKOS concept scheme and assigns a default tier to every property it defines, so that one record can be public at one property and elder-restricted at another.

The Contested Knowledge Gap

Afro-Atlantic sacred traditions are not monolithic. Lineage disputes, regional variation, diasporic divergence, and the ruptures of the Middle Passage have produced a knowledge landscape in which several authoritative perspectives on the same entity, practice, or text coexist without resolution. A framework that assigns a single authoritative value to a field cannot represent this without falsifying it. The choice of which lineage's version to record is itself an act of authority that the describing institution has no standing to make. The Iroko Framework treats contested knowledge as a normal condition: claims are recorded as assertions with their own authority and status, and each lineage's position can be held in its own named graph.

The Agency Gap

General-purpose description positions the institution as the actor and the described entity as passive. For living sacred traditions this is philosophically untenable, and for communities whose knowledge has been extracted and misrepresented by exactly this kind of description it is practically dangerous. The Iroko Framework models sacred agents, including spirits, practitioners, lineage authorities, and community governance bodies, as actors with authorization rights, refusal rights, and stewardship mandates. A spirit can be the asserted source of a restriction. A community council can be the body whose consent governs a record. The framework makes no metaphysical claim in doing so; it records how communities say their knowledge is governed.

The Custody Gap

The dominant archival model assumes that the institution holding materials is their custodian, with authority to describe them and decide about them. Postcustodial archival theory has questioned that assumption for more than four decades.[10] Community-centered initiatives have moved it into practice: the CARE Principles for Indigenous Data Governance set out collective benefit, authority to control, responsibility, and ethics as conditions for data use; Local Contexts gives communities Traditional Knowledge and Biocultural Labels and Notices that travel with collection records; and Mukurtu implements community-defined cultural protocols in a working platform.[11][12][13] The Iroko Framework builds on that work. Its particular contribution is to express custody as structure inside a domain ontology: access tiers bound to properties, authority bound to each assertion, and refusal recorded as an act, so that the governance logic travels with the data across linked data systems instead of residing in a single platform.

SECTION IV

The Framework as Binding

The Iroko Framework is not a metadata schema in the ordinary sense. It is the semantic infrastructure that makes three distinct institutional components legible to each other and to the outside world, and governable under one logic, without erasing what makes each component distinct.

An archive, a research library, and a living museum do not simply hold different collections. They stand in different relationships to knowledge. The archive's relationship is custodial and restricted. The library's is scholarly and mediated. The museum's is participatory and living. These are differences of kind, not degree. An infrastructure that flattens them in the name of interoperability produces a system that is technically functional and institutionally false.

The framework holds all three relationships through a single architectural decision: the separation of vocabulary publication from data publication. The vocabulary, sixteen modules, two alignment modules, 101 classes, 413 properties, 76 concept schemes, and 655 concepts, is public, CC0

dedicated, and open to adoption.[14] The records described with it are not public by default. Their disclosure is governed field by field according to the authorization frameworks of the traditions represented, and the infrastructure that enforces those decisions is specified in Section IX. Any institution can use the language. Only authorized parties can access what has been said in it.

This separation moves governance from policy documents into the descriptive model. Ethical description of Indigenous and sacred materials has long been pursued through principles and guidelines, and more recently through platforms and labels; the CARE Principles, Local Contexts, and Mukurtu have each advanced the field substantially.[15][16][17] The Iroko Framework adds a complementary layer. A restriction is not a note in a finding aid or a setting in one application. It is a property of the record, expressed in RDF at the level of the individual field, and traceable to the authority that established it. Whatever system later reads the record can read the governance with it.

The binding function operates at four levels.

At the level of the field and the passage, a single property value, or a single segment of a longer testimony, carries its own access level and its own provenance. This is the finest grain the framework supports today, and Section XII describes the conditions v2.0.0 is designed to add within it.

At the record level, one described entity, a sacred drum, an odu corpus, a lineage house, an interview turn, can carry properties drawn from several modules, each with its own access tier, provenance chain, and authorization context.

At the institutional level, the framework provides one descriptive language across three components that would otherwise need three systems with no reliable way to relate records across them.

At the community level, the framework makes sovereignty operational: communities and their authorities are encoded as governing agents whose authorization is recorded as the basis for asserting, modifying, or disclosing particular properties.

Remove the framework, and the archive, the library, and the museum become three institutions with separate systems, separate vocabularies, and no shared infrastructure for the question every one of them faces: who has the right to know what about the knowledge in their care, and on whose authority.

SECTION V

Architecture and Design Principles

The framework is organized into sixteen modules across three layers: a Foundation layer consisting of the Core module on which every other module depends; a Governance Layer of five composable modules that add sovereignty infrastructure to any domain application; and a Domain Layer of ten tradition-specific vocabularies. Two alignment modules relate the framework to PROV-O and Darwin Core. All terms share one flat namespace, `https://ontology.irokosociety.org/iroko#`, adopted in release v1.3.0; modules are separate documents, not separate namespaces.

The modular design is a response to institutional reality. Adopting institutions stand in different relationships to different knowledge domains, and none should have to implement governance infrastructure it does not need or use vocabulary it has no standing to use. Modules version independently, so their declared versions are deliberately uneven; the framework version is a release label (Appendix A.1).

Six design principles govern the architecture. Each emerged from a concrete institutional demand.

Field-Level Access Control

Six tiers, published as a concept scheme. Every property carries a default minimum tier; any record, field, or passage can be set higher. One entity carries public and elder-restricted values in the same record. PUBLISHED

Contested Knowledge Modeling

Assertion reification places each claim in a statement node recording its authority, its fieldwork event, and its status as consensual, lineage-specific, or contested. Lineage positions coexist in named graphs without forced resolution. PUBLISHED

Provenance Tracking

Oral transmission chains, fieldwork events, community endorsement, and epistemic type are first-class properties, aligned with PROV-O. The legitimacy of a description is inseparable from the legitimacy of the process that produced it. PUBLISHED

Sovereign Agency

Sacred agents, including spirits, practitioners, lineage authorities, and community bodies, are modeled as actors with authorization rights, refusal rights, and stewardship mandates. PUBLISHED

Postcustodial Design

The framework does not assume that the holding institution has authority to describe what it holds. Communities retain authority over their knowledge wherever it is held, and the model records whose authority each decision rests on. PUBLISHED

Semantic Interoperability

SKOS throughout; alignment modules for PROV-O and Darwin Core; FOAF, schema.org, Dublin Core Terms, and OntoLex-Lemon used in the modules that need them. The framework extends existing infrastructure rather than replacing it. PUBLISHED

Together these principles form one design philosophy: governance before description, sovereignty before access, community authority before institutional convenience. The technical decisions are downstream of the governance decisions, and the governance decisions are downstream of the communities whose knowledge the framework was built to protect.

SECTION VI

The Access Philosophy

The framework's most consequential decision is also the simplest to state: the vocabulary is public; the data is not. This is not a compromise between openness and restriction. It is a position about what institutions, scholars, and communities each have standing to know, and about the infrastructure required to make those distinctions operational.

The separation resolves a tension that has long troubled ethical description. Institutions holding Afro-Atlantic sacred materials have faced two bad options: describe everything openly, exposing knowledge communities have the right to restrict, or describe little of substance, producing finding aids too thin to be useful. The framework offers a third. Describe with full semantic richness. Govern access to each layer of that description independently. Publish the vocabulary so the description is interoperable. Protect the data so the knowledge remains under community authority.

ACCESS LEVEL CLASSIFICATION · IROKO:ACCESSLEVELSCHEME (SKOS NOTATION)

1	2	3	4	5	99
public unrestricted	public, no amplification	community only	initiated only	initiated elder	no access never exported

Five tiers govern who may view a value. The sixth, no access, is different in kind: it marks material that is operational only and is never exported to public RDF at all. Secret names are the clearest case; the property that holds them carries this tier by default. The elder tier is a socially recognized

designation. Nothing in the framework computes it, and nothing should; dated facts and eligibility evidence may support a community's recognition, but they do not replace it (Section XII).

The Grain of Disclosure

Access in the framework works at progressively finer grains. Four are available in release v1.4.0. A fifth, conditions within a tier, is the principal access change planned for v2.0.0.

1	**Property default** PUBLISHED Every property carries `iroko:minimumAccessLevel`, a floor published with the term. 426 terms carry one in release v1.4.0. Defaults let a system decide what is safe to show before anyone has reviewed a record.
2	**Record** PUBLISHED `iroko:accessLevel` raises the level for a whole record: one plant, one numbered item, one interview turn.
3	**Field** PUBLISHED An assertion about one property of one record carries its own access level, authority, and fieldwork event. An informant reference can be restricted while the citation beside it is public.
4	**Passage** PUBLISHED A testimony is divided into ordered `iroko:TextSegment` nodes, each with its own level, so one speaker's account can be partly open, partly restricted, and partly closed.
5	**Conditions within a tier** PLANNED, V2.0.0 Eligibility constraints scoped to tradition, office, gender, and dated eldership evidence; disclosure anchored to exact spans of a source page; refusal recorded outside ritual. Described in Section XII.

The framework accommodates four institutional relationships to this vocabulary, each legitimate and each governable under the same infrastructure. Universities and research institutions working mainly with published ethnographies and public holdings implement relatively open policies while respecting community tiers for restricted material. Temple archives and community institutions apply the same vocabulary with stricter calibrations, setting more fields to initiated and elder tiers. Museums may describe sacred objects at the public tier for exhibition while donor files or repatriation records require community authorization. Researchers in the field build provenance records that travel with field data into every later institutional context, so that governance obligations stay visible to every future user.

The Society's own practice shows the philosophy at work. The vocabulary and the public library are published openly on static hosting. The Ewé pilot publishes its fifty plant records openly, with never-exported material withheld at build. The Hyatt corpus is held entirely restricted by default. A correct first public release of Hyatt will show only an index of item numbers, printed pages, field sites, and subject terms, because no record has yet completed review. That is the system working as designed. Governed delivery of the reviewed material is specified in Section IX and will be built in the Hyatt pilot. DESIGNED

This is the access philosophy the framework embodies: not open access as a universal value, but governed access as a sovereignty right; not restriction as institutional caution, but restriction as community authority encoded in infrastructure. The framework does not ask institutions to trust communities. It builds community authority into the system so that trust is not the operative variable.

SECTION VII

The Modules: Built by Institutional Necessity

The sixteen modules were not designed speculatively. Each exists because the institution met a descriptive problem that no existing vocabulary solved and that the archive, library, or museum could not work without solving. All terms below share the `iroko:` namespace; module names identify the documents that define them.

Foundation

The Core module is the one module every other module depends on. It establishes the class hierarchy, the access level scheme, the assertion mechanism for contested knowledge, and the concept schemes shared across modules, among them the tradition, assertion status, and community endorsement schemes. Its nineteen classes include the foundational types required before any domain description is possible: `SacredEntity`, `LineageKnowledgeGraph`, `ContestedAuthority`, `DocumentaryEvidence`, `Assertion`, `RelationshipAssertion`, `RitualAuthority`, `TemporalVariation`, `AccessPolicy`, and `FieldworkEvent`. `Assertion` carries `assertionSubject`, `assertionPredicate`, and `assertionObject`, together with `authorizedBy`, `recordedInEvent`, `recordedOn`, and `assertionStatus`, so that any single claim, including a claim about who may see a field, has its own authority and provenance. Core also defines the never-exported properties, such as `secretName` and `invocationText`, whose values stay in the operational system.

Governance Layer

The five governance modules are composable. No institution must implement all of them; each declares its dependencies and degrades gracefully without the others. Together they form a sovereignty infrastructure that can be layered onto any domain module.

The **Agency** module addresses the absence, in general standards, of any mechanism for modeling sacred agents as actors. The traditions the Society documents do not separate human governance cleanly from sacred governance. Spirits authorize. Lineage authorities refuse. Divinatory offices transmit. The module models these through `SacredAgent`, `Spirit`, `RitualPractitioner`, `RitualEvent`, `AuthorizationEvent`, `RefusalEvent`, and `StewardshipMandate`, without claims about the ontological status of the entities involved. A `RefusalEvent` is a ritual event in which a sacred agent is asserted to refuse participation, disclosure, access, or a requested action; it is reached through `generatesRefusal` from the event, agent, or mandate that produced it. The framework does not assert that spirits exist in any particular sense. It asserts that communities govern their knowledge through structures that include sacred agents, and that honest infrastructure must represent those structures.

The **Authority** module develops the Core `RitualAuthority` into a full governance model: authority types, authority basis, jurisdiction, recognition networks, and temporal validity. Collections spanning several lineages, traditions, and periods need all of these, because each governance structure may recognize or contest the legitimacy of the others.

The **Epistemic** module governs knowledge gating: when, to whom, and under what conditions knowledge may be disclosed, withheld, or restricted. This is distinct from access control, which governs who may retrieve a record. Epistemic governance addresses the prior question of whether certain knowledge should be represented at all. `DisclosurePermission`, `DisclosureRestriction`, `SanctionedDisclosure`, and `ConstraintBasis` make that distinction operational.

The **Narrative** module builds full narrative modeling on Core's thin spine: narratives and their variants, testimonies, text segments, story transmission chains, contested kinship claims, and interpretive stances. A `Testimony` is a narrative variant with an identified speaker whose presence is itself part of the claim; `hasSegment` divides it into ordered `TextSegment` nodes. This is the module that carries passage-level governance for odu verse, pataki, interview testimony, and comparable corpora.

The **Manifestation** module models how sacred agents are asserted to manifest: embodiment, possession, dream, divination, symbolic presence, territorial claim, and auditory sign, with temporal variation where modes shift by ritual calendar. The living museum needs it because the difference between how an agent manifests in ceremony and how it is described in a record is a governance distinction the institution must represent.

Domain Modules

The ten domain modules cover distinct knowledge domains of Afro-Atlantic sacred practice. Each depends on Core, and each exists because an existing domain vocabulary was absent or could not carry the access and sovereignty requirements involved.

The **Ewé** module governs sacred plant knowledge, layering ritual governance over Darwin Core botanical description. Its published defaults show the field-level principle plainly. Specimen numbers, medicinal uses, contraindications, plant parts, and ritual uses are public. Ritual context, harvest protocol, preparation method, and constituents are community-only. Detailed preparation methods, preparation instructions, and divination sign references are initiated-only.

Contraindications are public by design: safety information should travel even when practice does not. Developed from Pierre Verger's ethnobotanical fieldwork corpus, the module underlies the Ewé pilot described in Section VIII.[18]

The **Nkisi** module covers spiritual entities across Afro-Atlantic traditions: orisha, lwa, mpungo, vodun, egun, Orí, and spirits of place, with camino and path modeling, syncretic identity tracking, spirit kinship, and polarity.

The **Travay** module distinguishes recurring ceremonies from threshold initiatory rites. That a ceremony exists is public. Its operational sequences and material details are community-only or initiated-only, so the living museum can describe its ceremonial calendar publicly while governing the knowledge that makes the ceremonies work.

The **Ilé** module covers religious houses, lineage, initiation genealogy, authority transmission, offices, and schism. Lineage is not background to the archive's materials; lineage is governance. Who initiated whom, through which house, determines the authority basis of every claim. `SchismEvent` addresses collections that document traditions in which disputes have produced competing genealogies.

The **Marca** module covers divination systems, sacred signs, readings, and verse corpora: Ifá with all 256 odu, Dilogún, Fá, Obi, and Chamalongo. Sign names are public. Verse texts and reading details are restricted. The module lets the corpus be described with scholarly richness while honoring the governance that Babaláwo communities have maintained over it.

The **Ékpè** module covers initiatory societies, graded initiation, esoteric governance, and masquerade: Ékpè and Ngbe, Abakuá, Ògbóni, Orò, Gẹ̀lẹ̀dẹ́, Egúngún, Poro, and Sande. It models lodges, grades, titles, membership, and masquerade traditions without exposing restricted grade content or member identities.

The **Vèvè** module covers graphic sign systems: vèvè, firmas, pontos riscados, anaforuana, nsibidi, adinkra, and Kongo cosmograms, with form, function, medium, surface, permanence, and lineage corpus.

The **Ngoma** module covers sacred music, rhythms, consecrated instruments, drum lineages, and musician authority transmission: Lucumí batá with Àñá, Vodou Rada and Petwo drumming, Candomblé atabaques, and Palo drumming. Consecrated instruments have their own governance, initiation genealogies, and transmission chains, and the module describes them as such.

The **Sankofa** module covers reclamation movements, diaspora returns, and reconstructed practice. Heritage relationships are modeled as assertions rather than facts, so multiple perspectives on the relation between a diaspora practice and its claimed source are documented without adjudicating authenticity.

The **Qal** module covers sacred lexicons, liturgical languages, and esoteric terminology. Its name is the Ge'ez word for "word," ቃል, and it reaches beyond the Caribbean Atlantic into Ethiopian and Eritrean manuscript traditions. It integrates OntoLex-Lemon for multilingual lexicography across

Lucumí, Kikongo, the sacred register of Haitian Kreyòl, Fon and Ewe, and Ge'ez. `SecretName` is a class in its own right, with its text held in a never-exported property, reflecting the reality that some names may not appear in any public record whatever the describer knows.

SECTION VIII

Two Governed Corpora: Ewé and Hyatt

A vocabulary earns trust in use. Two corpora now test the framework, and they test it at different grains. Ewé is a small, deep corpus of sacred plant knowledge in which governance attaches mainly to properties. Hyatt is a large corpus of recorded testimony in which governance must reach inside individual records. Together they show what release v1.4.0 already supports and what v2.0.0 needs to add.

Ewé: Property-Level Governance Proven Small

The Ewé pilot holds fifty plant records derived from Pierre Verger's fieldwork, with multilingual names in English, Spanish, French, Yorùbá, and Portuguese, and a Darwin Core model for botanical synonymy.[19] Its working tools implement the canonical access scheme directly. They evaluate each recorded use of a plant against its tier, decide which fields a given audience may see, and can redact public output accordingly. The defaults published in the Ewé module (Section VII) are the rules those tools apply. PUBLISHED

Ewé also taught the framework's most important operational lesson. Two pilot records tagged `access-no-access` in the source data were published and indexed anyway, before a never-export rule was added to the build, and for a time were reconciled by hand at each build. Nothing was wrong with the vocabulary. The failure was that enforcement depended on a person remembering. The conclusion carried into every later design is simple: the tier filter must be generated from the vocabulary and enforced in code, never by hand. What is not in the build output cannot be copied; everything else is friction.

The pilot also exposed a modeling question the next release must settle. Ewé holds folk names as alternative labels on a single plant node, which already requires hand-written notes wherever one folk name points to more than one botanical identification. Two nodes, a folk taxon and a botanical taxon, would let a tradition's own category stand as the primary object and carry plural, contested, and evidenced identifications. Section XII returns to this.

Hyatt: Governance Inside the Record

Between 1936 and 1942 the folklorist Harry Middleton Hyatt recorded interviews with hundreds of Black practitioners and informants across the American South. The material was published between 1970 and 1978 as *Hoodoo, Conjuration, Witchcraft, Rootwork*, five continuously paginated volumes

of numbered items and transcribed interviews.[20] The Society's extraction yields 20,279 records: 13,209 numbered items and 7,070 interview turns, with item numbering that reconstructs Hyatt's published count of 13,458 entries.[21]

The corpus maps onto the published vocabulary without new terms. An interview session is a `FieldworkEvent`, with its fieldworker, date, location, and protocol. A narrated item or interview turn is a `Testimony` divided into `TextSegment` nodes. An informant is an `ArchivalPerson`, identified by Hyatt's own informant number and deliberately left unresolved to any living person. An extracted claim is an `Assertion` recorded in its fieldwork event. Each volume is `DocumentaryEvidence` with a source citation, and disclosure is governed through `AccessPolicy`, `accessLevel`, and epistemic constraints. PUBLISHED

What Hyatt demands is grain. Descriptive metadata, source citations, informant references, and verbatim testimony carry different disclosure conditions, often within one interview turn. The illustration below follows the structure of one turn after review. Its values are synthetic, and no testimony is reproduced.

```
INTERVIEW TURN, AFTER REVIEW (illustrative; synthetic values)

  Volume, item number, printed page ........ public-unrestricted
  Field site (city, state) ................. public-unrestricted
  Subject terms (practice, materia) ........ public-unrestricted
  Informant number ......................... community-only
  Testimony, segment 1 ..................... community-only
  Testimony, segment 2 ..................... initiated-only
  Question the speaker declined ............ recorded as declined; no text exists

  Every level is an assertion recording who set it, on what
  authority, and why. Any field not yet reviewed: restricted.
```

A public user can find and cite the turn: where it sits in the volumes, where it was recorded, what it concerns. A verified community member can read the first segment of testimony; an initiate, both. Nothing is visible before review, which is why a correct first public release of the corpus shows only its index.

Three arguments converge on this shape, and their convergence is the design's strongest support. The governance argument holds that the verbatim testimony of Black practitioners, recorded under Jim Crow by a white folklorist for a scholarly readership, should not become freely redistributable a second time. The copyright argument holds that the volumes, published between 1970 and 1978, remain in copyright, so that republishing thousands of verbatim entries would reproduce a substantial portion of a protected work. The infrastructure argument holds that the contribution worth funding is the structured layer, a transformation of form rather than a reproduction of content. All three point to the same build: structured metadata public after review, verbatim testimony governed. Rights status is recorded separately from cultural disclosure, in Dublin Core Terms and RightsStatements.org values, so that a cultural decision is never mistaken for copyright clearance.[22]

Hyatt also clarifies what the system controls. Access to a record, authorization to practice, authority to teach, and license to sell are four different gates, and much public argument about Hoodoo treats them as one. The framework governs the first. It records who decided each disclosure, within which tradition, and on what standing. A decision-maker acts only within the traditions and relationships in which he or she holds standing, or through a designee appointed in writing for a defined corpus, tier, and tradition scope; each decision is logged with its rationale; and where no one with standing has decided, the material remains restricted pending the tradition's own custodians. Holding an administrative account confers no cultural decision authority.

Finally, Hyatt is where the vocabulary met its limits in useful ways. The mapping produced a register of needs for the next release rather than ad hoc local terms: refusal outside ritual, rights kept distinct from access, curator review kept distinct from community endorsement, durable anchors from a record to an exact span of a scanned page, and a tradition hierarchy that does not misfile Hoodoo under a single conflated parent. Each is taken up in Section XII.

SECTION IX

Reference Architecture for Governed Delivery

An access level in RDF is a statement, not a lock. The vocabulary declares who may see each field; an application must enforce it. This section specifies that application as the Society has designed it for the Hyatt corpus. It is written as a portable design with a reference deployment, so that any adopting institution can implement the same pattern on infrastructure it already uses. DESIGNED

Two Layers, One Boundary

The open layer, the ontology, the public research library, and the site shell, remains on static hosting, as it is today. Static hosting cannot authenticate a user, and that limitation marks exactly the boundary the governance requires. Governed records live in a second layer built beside the first: an application that authenticates each user and decides, field by field, what to return. Nothing in the open layer moves. The static pipeline publishes no unreviewed governed record.

Component	Design	Reference deployment
Canonical store	One PostgreSQL database holding every record, with stable identifiers that survive correction, splitting, and merging, a history of changes, and review status restricted by default.	Amazon RDS for PostgreSQL with automated backups
Application	Next.js interface for search, record views, citations, and the staff workflow; individual accounts through Auth.js.	Managed application hosting suited to Next.js
Policy enforcement	An API that checks every request against the requester's tier and applies approved field and passage decisions on the server. Restricted content is never sent to the browser to be hidden there.	Within the application tier

Component	Design	Reference deployment
Source images and files	Object storage reached only through defined roles; files checksummed when migrated from working folders so each can be matched to its original.	Amazon S3 with AWS Identity and Access Management
Monitoring and recovery	Alerts on application errors and failed access attempts; restoration from backup demonstrated before launch.	Amazon CloudWatch
Exports	Filtered RDF and machine-readable downloads generated from the one canonical dataset, containing approved fields only, with documented permissions, pagination, and limits. No unrestricted bulk export.	Generated by the application

Roles and Authority

The role model runs Anonymous, Public Registered, Community, Initiated, Initiated Elder, Steward, and Administrator. The first five correspond to viewing tiers; the last two are operational. The distinction that matters most is between technical and cultural authority. An administrator can manage accounts and restore a backup. An administrator cannot, by virtue of that role, decide what a tradition discloses. Sponsorship or payment never maps above Public Registered.

From Vocabulary to Enforcement

The lesson of the Ewé build governs this design: rules maintained by hand drift. The 426 published default levels, together with the record, field, and passage decisions stored in the canonical database, generate the permissions matrix. Local working labels are normalized to the canonical identifiers before any export or enforcement. The same matrix generates the acceptance tests: requests from an anonymous user and from test accounts at every tier, across search results, record views, downloads, and withdrawal. A field withdrawn after publication must disappear from all four. Search results, record views, and downloads apply the same decisions, so no route returns what another route withholds. A retrieval pipeline for AI systems built on the same service inherits the same checks, which matters because a language model cannot unlearn what reaches its context.

The Review-to-Publication Workflow

A restricted staff workflow lets the curator and authorized decision-makers compare each record with its source pages, record corrections with their history, document disclosure decisions with their authority and rationale, approve individual fields, and publish or withdraw approved material. Copyright review is recorded separately from cultural review. A completed review may end in public clearance, partial clearance, or continued restriction; continued restriction is a valid outcome, not a failure.

The Pilot

The first implementation is bounded deliberately. Over twelve months, all 20,279 Hyatt records enter the canonical database restricted by default; 321 records, drawn from all five volumes and from both numbered items and interview turns, receive full source, citation, rights, and field-level disclosure review; the access, workflow, and recovery tests pass before a controlled user pilot; and approved fields launch as a public collection guide and searchable discovery service with stable citations. The pilot runs on release v1.4.0 as published. It requires no new ontology development, which keeps the v2.0.0 vocabulary work on its own timeline while the pilot supplies the evidence that work will use.

SECTION X

The Institutional Landscape

The Iroko Framework was built to serve the Iroko Historical Society. It is published to serve a field. The institutions that hold Afro-Atlantic sacred materials are numerous and diverse, and most are under-equipped to describe, govern, and provide access to those materials on terms their source communities would accept. The shortage is rarely goodwill. Major universities, national archives, and museums have made public commitments to ethical stewardship of Indigenous and diaspora collections. The shortage is infrastructure: commitments made in policy cannot be honored in operation without vocabulary, governance mechanisms, and access architecture designed for the materials in question.

INSTITUTION TYPE	PRIMARY NEED	KEY MODULES
HBCUs with African diaspora collections	Descriptive vocabulary for sacred and restricted materials accumulated over a century without adequate schema	Core, Ilé, Nkisi, Travay, Narrative
Caribbean national archives	Governance infrastructure for Afro-religious community documentation held under colonial and republican administrative frameworks	Ékpè, Marca, Agency, Authority
Museum repatriation programs	Provenance, community governance context, sacred function, and authority transmission that collection management systems cannot record	Nkisi, Travay, Agency, Authority, Ngoma
Digital humanities centers	Standards-aligned domain vocabulary and a governed delivery pattern for corpora of recorded testimony	Core, Narrative, Epistemic; Section IX
Community-based archives	Interoperability without compromise of community governance authority	Core, Epistemic, Agency, access scheme
Theological seminaries	Lexicographic and liturgical infrastructure for African and diaspora sacred language collections	Qal, Marca, Narrative

Historically Black Colleges and Universities with African diaspora collections are an urgent adoption case. Institutions such as Howard University, Spelman College, Fisk University, and Dillard University hold Afro-Atlantic religious, cultural, and historical materials gathered over more than a century. Some have never been fully described; others have been described with general-purpose schemas that flatten their sacred and governance dimensions. The framework gives them a vocabulary designed for these materials.

Caribbean national archives face a version of the challenge shaped by their own histories. Cuba's archives hold extensive documentation of Afro-Cuban religious communities, cabildos, and initiatory societies created under colonial, republican, and revolutionary administrations. Haiti's archives hold materials whose sacred dimensions are inseparable from their historical significance. Trinidad and Tobago, Jamaica, Brazil, and Colombia each hold national collections that include Afro-Atlantic sacred materials described under frameworks that serve neither scholars nor communities well.

Museum repatriation programs meet the same problem from another direction. Institutions committed to returning materials often cannot describe them in terms communities would recognize as accurate or respectful. Repatriation requires documentation of provenance, governance context, sacred function, and authority transmission that general collection management systems do not hold.

Digital humanities centers working with recorded testimony, from folklore collections to oral history projects, face the problem Hyatt makes concrete: a corpus too valuable to leave undescribed and too sensitive to publish whole. The governed delivery pattern in Section IX is designed to be reused by exactly these projects.

Theological seminaries and religious studies programs with African and diaspora collections meet the framework through Marca, Qal, and Ékpè. The Ifá corpus, the Vodou liturgical tradition, and the governance of initiatory societies have drawn serious scholarship from institutions whose descriptive frameworks grew up around Christian and Near Eastern texts. Qal's accommodation of Ge'ez alongside Lucumí and Kikongo signals the framework's reach to institutions working where Afro-Atlantic and African Christian manuscript traditions meet.

The Society is the demonstration case across this landscape. It was built with the framework as its binding infrastructure, and the Ewé and Hyatt corpora are the first public evidence of how the framework performs. Institutions considering adoption will be able to examine the Hyatt pilot directly: its permissions matrix, its test results, its review files, and its operating costs.

SECTION XI

What Support Enables

The vocabulary is published, versioned, and open to adoption. The design for governed delivery is complete. What does not yet exist is the operating infrastructure, the community partnership network, and the institutional capacity to carry the framework from a published vocabulary into everyday use across the institutions that need it. That is what support enables, in three phases.

I Governed Delivery and Community Authorization Protocols

The first phase builds the reference architecture of Section IX and proves it on a bounded corpus. The Hyatt pilot places all 20,279 records in one canonical database restricted by default, completes source, citation, rights, and field-level disclosure review for 321 records across all five volumes, passes access, workflow, and recovery tests for every tier, and launches a public collection guide and discovery service for the approved fields. Its deliverables are reusable by any adopting institution: the permissions matrix, the acceptance tests, the review protocol, the measured cost of review per record, and a documented operating plan.

Community authorization protocol development runs alongside it and is the most demanding component. The framework provides the infrastructure for community governance; it does not prescribe the protocols by which particular communities govern their knowledge. Those protocols must be developed with the communities themselves, documented in forms the framework's `AccessPolicy`, `StewardshipMandate`, and authority structures can carry, and tested against real collections. The Hyatt pilot's written delegation of decision authority, scoped by corpus, tier, and tradition, is the first such protocol in operation.

An implementation guide tailored to four institutional types, research universities, community-based archives, museums with repatriation programs, and digital humanities centers, completes the phase. Each has a different technical environment, a different relationship to source communities, and different governance requirements; one generic guide would serve none of them well.

II Fieldwork, Validation, and Archivist Training

The second phase extends the framework's reach and validates its descriptive adequacy against knowledge domains and governance structures not yet fully represented. Fieldwork in Nigeria, Benin, Cuba, Haiti, Brazil, Mexico, and Colombia, conducted with community governance authorities, will test existing modules against new cases and supply the evidence for the domain extensions the version roadmap anticipates.

Archivist training has the most immediate institutional effect. Adoption depends on archivists who can implement linked data systems and who understand the cultural and governance dimensions that make these materials unlike other collections. Technical competence without governance competence produces

implementations that use the vocabulary correctly and apply it wrongly. A fellowship program for emerging archivists from Afro-Atlantic communities, combining technical training, governance education, and practical work at the Society, addresses the longer-term workforce need.

III Full Operational Capacity

The third phase is the Society operating at full capacity across all three components, with the framework governing the archive, library, and museum in production.

For the archive, that means governed collections representing the breadth of the Society's fieldwork holdings across Cuba, Nigeria, Benin, and the wider Afro-Atlantic world, with community authorization protocols in place for every restricted collection, and governed access extended across a network of affiliated institutions. Federated access requires authentication that can recognize community authorization across institutional boundaries, query interfaces that respect field-level restrictions whichever institution is asking, and audit trails that make the governance history of every record visible to the communities whose authorization governs it.

For the library, it means a curated and interoperable collection of published ethnographies, digitized historical materials, and community-authorized research, linked through the vocabulary to related materials at affiliated institutions.

For the living museum, it means an institutional presence adequate to its mandate: physical and digital exhibition that makes Afro-Atlantic sacred traditions visible in forms that honor their living complexity, educational programming developed with the communities represented, and curatorial practice governed by the framework's sovereignty architecture.

One principle runs through all three phases: the communities whose knowledge is at stake are not the beneficiaries of this work. They are its governing authorities. Support for the framework is support for infrastructure that lets them exercise that authority in the systems that hold their knowledge.

SECTION XII

What the Corpora Teach: Granularity in v2.0.0

Release v1.4.0 governs disclosure down to the property, the record, the field, and the passage. The corpora show where the next grain lies. The planned v2.0.0 release adds conditions within a tier: not only whether a viewer is initiated, but within which tradition, for which office, and on what evidence of eldership; not only which passage, but which span of which page; not only ritual refusal, but any recorded refusal to disclose. PLANNED, V2.0.0

Work on v2.0.0 is paused while the governed delivery pilot proceeds, and its scope is not yet fixed. Architecture decision records for the first two changes are held on a development branch; no published term has changed. Two commitments already hold regardless of scope: the Core module moves to 2.0.0 because the tradition change is breaking, and no published URI is ever deleted, only deprecated in place with `dcterms:isReplacedBy` pointing to its successor.

Area	Release v1.4.0	Direction for v2.0.0
Tradition	One SKOS hierarchy in which a single broader relation carries four different relations: subsumption, ethnolinguistic substrate, compositional or political grouping, and geography. Hoodoo is consequently misfiled at the top-level rollup across all 13,209 numbered Hyatt items.	Four separate axes, so that a disclosure condition can be scoped to a tradition without inheriting a misleading parent.
Eldership	`access-initiated-elder` is a socially recognized designation. Nothing records the dated facts a community may point to.	Dated birth and initiation, with the framework's existing precision and uncertainty apparatus, so that natal and initiatory seniority can be derived at query time. The tier itself remains recognized, never computed.
Eligibility	Gender appears in three uncoordinated places: one constraint-basis concept and two free-text properties scoped to particular module domains.	One general, queryable constraint scoped to tradition and office, distinguishing who may know from who may hold or perform a role. Whether this extends `EpistemicConstraint` or adds a sibling class is an open decision.
Refusal	`RefusalEvent` is a ritual event involving a sacred agent.	A broader refusal to disclose, covering an interviewee who declines a question, with `RefusalEvent` as its ritual case. Prompted by Hyatt.
Source anchors	Citations to volume, item, and printed page.	Anchors from a record or segment to an exact span of a scanned page, evaluated against the W3C Web Annotation Data Model, so that disclosure can follow the source as closely as the description does.[23]
Plant identity	Folk names held as alternative labels on one botanical node.	A folk taxon and a botanical taxon as separate nodes, so one folk name can carry plural, contested, evidenced identifications. Prompted by Ewé.
Review and endorsement	Community endorsement is modeled; curator review is not distinguished from it.	A small workflow scheme for technical review, kept separate from community endorsement so that one is never read as the other.

Area	Release v1.4.0	Direction for v2.0.0
Typing by tier	Class membership is public wherever a record is visible.	An access-stratified typing mechanism, under consideration, so that what kind of thing an entity is can itself be disclosed by tier.

Two things stay outside the vocabulary by design. Copyright and licensing remain in Dublin Core Terms and RightsStatements.org rather than in Iroko tiers, because a legal status and a cultural decision answer different questions.[24] Extraction and transformation provenance remain in PROV-O, which already owns that problem. The framework adds terms only where governance or query needs cannot be met by an existing standard.

The sequence matters. The Hyatt pilot runs on release v1.4.0 and produces evidence: which conditions reviewers actually needed, which refusals they recorded, which spans they had to anchor. That evidence is what v2.0.0 should be built from.

SECTION XIII

Conclusion

Afro-Atlantic sacred knowledge systems have survived the Middle Passage, colonial suppression, institutional neglect, and a century of extraction by scholars and institutions that described them without governing them and preserved them without serving the communities that held them. That survival is not incidental. It is the result of deliberate, community-governed strategies for keeping authority over knowledge under conditions designed to remove it. The traditions are not fragile. The infrastructure to steward them in the digital age has been.

The Iroko Historical Society and the Iroko Framework are built to supply that infrastructure from the inside out: a postcustodial complex designed by practitioners with standing in the traditions it serves, governed by the communities with authority over their knowledge, and bound by a vocabulary able to carry that authority down to a single field or a single passage of testimony. The vocabulary is published and standards-aligned. Two governed corpora have tested it. The application that will enforce it is fully specified.

The unmet need is concrete. There are sacred drums in university storage described as percussion instruments. There are initiatory lineage documents in national archives filed as miscellaneous correspondence. There are odu verse collections in special collections catalogued as folklore. There are thousands of pages of recorded testimony that can be either published whole or left undescribed, because no system has offered a governed middle. The materials survived. The description failed them.

The next proof is a bounded one: one corpus, one canonical database, every field reviewed and every decision recorded, enforced in code and tested at every tier. What the field requires is the will to adopt infrastructure that centers community sovereignty rather than institutional convenience, and the investment to build it first where it can be tested and then where it is needed most.

Notes

1. Iroko Historical Society, *Iroko Framework* vocabulary, release v1.4.0, eighteen Turtle files, https://ontology.irokosociety.org/vocab/; source repository https://github.com/iroko-framework/iroko-framework, commit 57503f7. Counts computed by parsing all eighteen files, October 20, 2026. ↩
2. Pierre Fatumbi Verger, *Ewé: The Use of Plants in Yoruba Society* (São Paulo: Odebrecht, 1995). The pilot database of fifty records is at https://ewe-db.irokosociety.org. ↩
3. Harry Middleton Hyatt, *Hoodoo, Conjuration, Witchcraft, Rootwork*, 5 vols. (Hannibal, MO: Western Publishing, for the Alma Egan Hyatt Foundation, 1970-1978). Fieldwork dates and the terms on which informants were interviewed: vol. 1, `pinpoint TK`. ↩
4. Kimberly Christen, Alex Merrill, and Michael Wynne, "A Community of Relations: Mukurtu Hubs and Spokes," *D-Lib Magazine* 23, no. 5/6 (May/June 2017), secs. 1 and 3, https://doi.org/10.1045/may2017-christen. ↩
5. On terreiro-held museums, see for example the museum of Ilê Axé Opô Afonjá in Salvador, Bahia: "Museu do Ilê Opô Afonjá terá estrutura e acervo sagrado restaurados," *Bahia Notícias*, https://www.bahianoticias.com.br/noticia/295285-museu-do-ile-opo-afonja-tera-estrutura-e-acervo-sagrado-restaurados `verify date and details before publication`. ↩
6. Dublin Core Metadata Initiative, "Dublin Core Metadata Element Set, Version 1.1: Reference Description," https://www.dublincore.org/specifications/dublin-core/dces/. ↩
7. DCMI Usage Board, "DCMI Metadata Terms," January 20, 2020, s.v. "Access Rights," https://www.dublincore.org/specifications/dublin-core/dcmi-terms/2020-01-20/. ↩
8. Library of Congress, "<accessrestrict> Conditions Governing Access," in *Encoded Archival Description Tag Library, Version 2002*, https://www.loc.gov/ead/tglib/elements/accessrestrict.html. The element may occur within `<archdesc>` and within every component level from `<c>` and `<c01>` through `<c12>`. ↩
9. Renato Iannella and Serena Villata, eds., *ODRL Information Model 2.2*, W3C Recommendation, February 15, 2018, sec. 1.3, https://www.w3.org/TR/odrl-model/. ↩
10. F. Gerald Ham, "Archival Strategies for the Post-Custodial Era," *American Archivist* 44, no. 3 (1981): 207-16. ↩
11. Stephanie Russo Carroll et al., "The CARE Principles for Indigenous Data Governance," *Data Science Journal* 19, no. 1 (2020): 43, https://doi.org/10.5334/dsj-2020-043. ↩
12. Local Contexts, "Traditional Knowledge and Biocultural Labels and Notices," accessed October 20, 2026, https://localcontexts.org/; DataCite, "Local Contexts Notices and Labels," DataCite Support, https://support.datacite.org/docs/local-contexts-notices-and-labels. ↩
13. Christen, Merrill, and Wynne, "Community of Relations," sec. 3. ↩
14. Iroko Historical Society, *Iroko Framework* vocabulary, release v1.4.0. ↩
15. Carroll et al., "CARE Principles." ↩
16. Local Contexts, "Labels and Notices." ↩
17. Christen, Merrill, and Wynne, "Community of Relations," sec. 3. ↩
18. Verger, *Ewé*. ↩
19. Verger, *Ewé*. ↩
20. Hyatt, *Hoodoo, Conjuration, Witchcraft, Rootwork*. ↩
21. Iroko Historical Society, Hyatt corpus extraction inventory and ontology mapping, Per Medjat project records, September 2026 (unpublished). ↩
22. RightsStatements.org, "Rights Statements," https://rightsstatements.org/; DCMI Usage Board, "DCMI Metadata Terms," s.v. "Rights," "Rights Holder," and "License." ↩
23. Robert Sanderson, Paolo Ciccarese, and Benjamin Young, eds., *Web Annotation Data Model*, W3C Recommendation, February 23, 2017, https://www.w3.org/TR/annotation-model/. ↩
24. RightsStatements.org, "Rights Statements." ↩

APPENDIX A

Technical Reference

A.1 Module Index

Counts are computed from the eighteen published vocabulary files of release v1.4.0. Versions are each module's declared `owl:versionInfo`; modules version independently.

MODULE	DOCUMENT	LAYER	VERSION	CLASSES	PROPERTIES	SCHEMES	CONCEPTS
Core	iroko-core	Foundation	1.4.0	19	80	11	117
Agency	iroko-agency	Governance	1.3.0	11	11	0	0
Authority	iroko-authority	Governance	1.3.1	6	12	4	31
Epistemic	iroko-epistemic	Governance	1.3.0	5	5	1	8
Narrative	iroko-narrative	Governance	1.3.0	5	24	5	33
Manifestation	iroko-manifestation	Governance	1.3.1	5	9	3	24
Ewé	iroko-ewe	Domain	1.4.0	4	23	5	62
Nkisi	iroko-nkisi	Domain	1.3.0	4	28	6	54
Travay	iroko-travay	Domain	1.3.0	4	22	5	41
Ilé	iroko-ile	Domain	1.3.0	6	38	5	58
Marca	iroko-marca	Domain	1.3.0	5	25	4	32
Ékpè	iroko-ekpe	Domain	1.3.0	7	37	5	34
Vèvè	iroko-veve	Domain	1.3.0	4	14	5	44
Ngoma	iroko-ngoma	Domain	1.3.0	5	27	4	31
Sankofa	iroko-sankofa	Domain	1.3.1	5	31	8	49
Qal ቃል	iroko-qal	Domain	1.3.0	6	27	5	37
PROV-O alignment	iroko-align-prov	Alignment	1.3.0	0	0	0	0
Darwin Core alignment	iroko-align-dwc	Alignment	1.0.0	0	0	0	0
Totals			**Release v1.4.0**	**101**	**413**	**76**	**655**

A.2 Namespace Declarations

TURTLE PREFIX DECLARATIONS

```
# One flat namespace for every module since release v1.3.0
@prefix iroko:   <https://ontology.irokosociety.org/iroko#> .
# Module documents are identified by rdfs:isDefinedBy, for example
#   <https://ontology.irokosociety.org/iroko-core>
#   <https://ontology.irokosociety.org/iroko-ewe>
# External vocabularies used in the examples
@prefix skos:    <http://www.w3.org/2004/02/skos/core#> .
@prefix prov:    <http://www.w3.org/ns/prov#> .
@prefix dcterms: <http://purl.org/dc/terms/> .
@prefix dwc:     <http://rs.tdwg.org/dwc/terms/> .
@prefix xsd:     <http://www.w3.org/2001/XMLSchema#> .
```

A.3 Standards Alignment

EXTERNAL STANDARD	WHERE USED	RELATIONSHIP
SKOS	All modules	Concept scheme structure throughout, including the access level scheme
PROV-O	Alignment module; Core, Agency, Narrative, Ilé, Ékpè	Assertions, access policies, lineage graphs, and documentary evidence as `prov:Entity`; ritual and fieldwork events as `prov:Activity`; sacred agents and ritual authorities as `prov:Agent`
Darwin Core	Alignment module; Ewé	Botanical specimen and occurrence data; the plant class alignment is under review (Appendix D)
Dublin Core Terms	All modules	Module metadata, licensing, bibliographic resources for source citation; rights status for governed corpora
FOAF	Core, Ilé, Ékpè, Marca, Ngoma, Sankofa, Vèvè	Persons as fieldworkers, initiates, office holders, and witnesses
schema.org	Ilé, Ékpè, Sankofa	Religious houses and societies as organizations
OntoLex-Lemon	Qal	Multilingual lexicographic infrastructure
W3C Web Annotation	Planned, v2.0.0	Record and segment anchors to spans of scanned pages

A.4 Access Level Definitions

NOTATION	IDENTIFIER	DEFINITION (IROKO:ACCESSLEVELSCHEME)
1	`iroko:access-public-unrestricted`	Freely accessible to anyone. No authentication required.
2	`iroko:access-public-no-amplification`	Publicly viewable but should not be amplified, shared widely, or used for commercial purposes.
3	`iroko:access-community-only`	Accessible to verified community members. Requires authentication.
4	`iroko:access-initiated-only`	Accessible only to initiated practitioners whose initiation has been verified.
5	`iroko:access-initiated-elder`	Requires elder designation within the tradition, recognized by the community as having authority to access lineage-specific teachings.
99	`iroko:access-no-access`	Operational or internal use only. Never exported to RDF.

APPENDIX B

Sample RDF

Examples B.1 to B.4 use only terms published in release v1.4.0. Instance identifiers use an example namespace, and every value is synthetic. Example B.5 shows the planned direction and is marked as such; its term names are placeholders, not proposals.

B.1 Property Defaults and a Field Raised: Ewé

The vocabulary publishes a default level with each property. A community can raise one field above its default for one plant through an assertion that records its own authority.

TURTLE · EWÉ PROPERTY DEFAULTS AND A FIELD-LEVEL ASSERTION

```
@prefix iroko: <https://ontology.irokosociety.org/iroko#> .
@prefix ex:    <https://example.org/ewe/> .
@prefix xsd:   <http://www.w3.org/2001/XMLSchema#> .

# Published vocabulary (excerpt): defaults travel with each property
iroko:contraindication          iroko:minimumAccessLevel iroko:access-public-unrestricted .
iroko:ritualContext             iroko:minimumAccessLevel iroko:access-community-only .
iroko:preparationMethodDetailed iroko:minimumAccessLevel iroko:access-initiated-only .
iroko:secretName                iroko:minimumAccessLevel iroko:access-no-access .

# Instance data: one plant, values withheld in this example
ex:plant-0001 a iroko:Plant ;
    iroko:contraindication          "[public safety note]" ;
    iroko:ritualContext             "[community-only context]" ;
    iroko:preparationMethodDetailed "[initiated-only method]" .

# The holding house raises ritual context above its default for this plant
ex:assertion-0001-context a iroko:Assertion ;
    iroko:assertionSubject   ex:plant-0001 ;
    iroko:assertionPredicate iroko:ritualContext ;
    iroko:accessLevel        iroko:access-initiated-elder ;
    iroko:authorizedBy       ex:authority-house-a ;
    iroko:recordedInEvent    ex:fieldwork-0001 ;
    iroko:recordedOn         "2026-03-02"^^xsd:date .

ex:authority-house-a a iroko:RitualAuthority .
ex:fieldwork-0001    a iroko:FieldworkEvent .
```

B.2 Field and Passage Governance: a Hyatt Interview Turn

The citation and field site stay at their public defaults. The informant reference is restricted by a field-level assertion. The testimony is divided into segments, each with its own level. No testimony text is reproduced.

TURTLE · HYATT TESTIMONY WITH FIELD AND SEGMENT LEVELS

```
@prefix iroko:   <https://ontology.irokosociety.org/iroko#> .
@prefix dcterms: <http://purl.org/dc/terms/> .
@prefix ex:      <https://example.org/hyatt/> .
@prefix xsd:     <http://www.w3.org/2001/XMLSchema#> .

ex:session-v0-i00 a iroko:FieldworkEvent ;
    iroko:eventLocation "[city, state]"@en ;
    iroko:eventDate     "1938-01-01"^^xsd:date .   # synthetic

ex:volume-0 a iroko:DocumentaryEvidence , dcterms:BibliographicResource .
ex:informant-0000 a iroko:ArchivalPerson .   # never resolved to a living person

ex:turn-v0-i00-t0000 a iroko:Testimony ;
    iroko:tradition         iroko:tradition-hoodoo ;
    iroko:hasSourceCitation ex:volume-0 ;
    iroko:narratedBy        ex:informant-0000 ;
    iroko:hasSegment        ex:turn-v0-i00-t0000-s1 , ex:turn-v0-i00-t0000-s2 .

# Passage level: each segment carries its own decision
ex:turn-v0-i00-t0000-s1 a iroko:TextSegment ;
    iroko:segmentOrder 1 ;
    iroko:accessLevel  iroko:access-community-only .
ex:turn-v0-i00-t0000-s2 a iroko:TextSegment ;
    iroko:segmentOrder 2 ;
    iroko:accessLevel  iroko:access-initiated-only .

# Field level: the informant reference is restricted; the citation is not
ex:decision-t0000-narrator a iroko:Assertion ;
    iroko:assertionSubject   ex:turn-v0-i00-t0000 ;
    iroko:assertionPredicate iroko:narratedBy ;
    iroko:assertionObject    ex:informant-0000 ;
    iroko:accessLevel        iroko:access-community-only ;
    iroko:authorizedBy       ex:designee-hoodoo ;
    iroko:recordedOn         "2027-01-15"^^xsd:date .

ex:designee-hoodoo a iroko:RitualAuthority .   # scoped in writing by corpus, tier, tradition
# A declined question is held as a project flag in v1.4.0; see B.5.
```

B.3 Contested Knowledge: Two Lineage Positions

Two houses make incompatible claims about the transmission of an office. Both claims are recorded with their authority, fieldwork source, and status. Neither is adjudicated.

TURTLE · ASSERTION REIFICATION FOR A CONTESTED CLAIM

```
@prefix iroko: <https://ontology.irokosociety.org/iroko#> .
@prefix ex:    <https://example.org/ile/> .

ex:office-x a iroko:ReligiousOffice ;
    iroko:accessLevel    iroko:access-community-only ;
    iroko:competingClaim ex:claim-house-a , ex:claim-house-b .

ex:claim-house-a a iroko:Assertion ;
    iroko:assertionSubject   ex:office-x ;
    iroko:assertionPredicate iroko:grantedBy ;
    iroko:assertionObject    ex:office-held-in-house-a ;
    iroko:assertionStatus    iroko:assertion-contested ;
    iroko:authorizedBy       ex:authority-house-a ;
    iroko:recordedInEvent    ex:fieldwork-a ;
    iroko:supportingLineages "House A and its branches" ;
    iroko:rejectingLineages  "House B" ;
    iroko:accessLevel        iroko:access-community-only .

ex:claim-house-b a iroko:Assertion ;
    iroko:assertionSubject   ex:office-x ;
    iroko:assertionPredicate iroko:grantedBy ;
    iroko:assertionObject    ex:office-held-in-house-b ;
    iroko:assertionStatus    iroko:assertion-contested ;
    iroko:authorizedBy       ex:authority-house-b ;
    iroko:recordedInEvent    ex:fieldwork-b ;
    iroko:supportingLineages "House B" ;
    iroko:rejectingLineages  "House A and its branches" ;
    iroko:accessLevel        iroko:access-community-only .

# Each house's full position may also be held in its own named graph
ex:graph-house-a a iroko:LineageKnowledgeGraph .
ex:graph-house-b a iroko:LineageKnowledgeGraph .
```

B.4 Sovereign Agency: a Refusal Event

The refusal is public. What was refused appears nowhere in the system. An institution using the framework cannot accidentally expose what was explicitly declined.

TURTLE · IROKO:REFUSALEVENT

```
@prefix iroko: <https://ontology.irokosociety.org/iroko#> .
@prefix prov:  <http://www.w3.org/ns/prov#> .
@prefix ex:    <https://example.org/ngoma/> .
@prefix xsd:   <http://www.w3.org/2001/XMLSchema#> .

ex:ceremony-0001 a iroko:RitualEvent ;
    iroko:generatesRefusal ex:refusal-0001 .

ex:refusal-0001 a iroko:RefusalEvent ;
    prov:startedAtTime   "2019-11-03T00:00:00"^^xsd:dateTime ;
    iroko:involvesAgent  ex:presiding-drummer ;
    iroko:accessLevel    iroko:access-public-unrestricted .

ex:presiding-drummer a iroko:RitualPractitioner .
# The declined consecration protocol is not represented in any record.
```

B.5 The Planned Grain PLANNED, V2.0.0

Not part of release v1.4.0. Term names in the `planned:` namespace are placeholders that show the shape of the change; the decisions behind them remain open.

TURTLE · ILLUSTRATIVE ONLY, NOT PUBLISHED VOCABULARY

```
@prefix iroko:   <https://ontology.irokosociety.org/iroko#> .
@prefix planned: <https://example.org/planned/> .
@prefix ex:      <https://example.org/hyatt/> .

# Refusal outside ritual: a speaker declines a question
ex:declined-t0000-q3 a planned:DisclosureRefusal ;
    iroko:involvesAgent ex:informant-0000 ;
    iroko:accessLevel   iroko:access-public-unrestricted .

# A condition within a tier: initiated, within one tradition, for one office
ex:segment-condition a planned:EligibilityConstraint ;
    planned:appliesTo     ex:turn-v0-i00-t0000-s2 ;
    planned:withinTier    iroko:access-initiated-only ;
    planned:tradition     iroko:tradition-hoodoo ;
    planned:eligibleRole  ex:office-z .

# A segment anchored to an exact span of its scanned source page
ex:turn-v0-i00-t0000-s2 planned:anchoredBy ex:annotation-0001 .   # a Web Annotation
```

APPENDIX C

Institutional Architecture Diagram

Read top to bottom: community authority as the governing source, flowing through the framework's three module layers and the access scheme, through the governed delivery layer that enforces it, into the three institutional components, and down to the standards that keep the whole interoperable.



APPENDIX D

Changes from the First Edition

Item	First edition	Second edition
Counts	Three inconsistent sets: 91 classes and 379 properties; 93 and 386; 94 and 389	101 classes, 413 properties, 76 schemes, 655 concepts, computed from the files
Access tiers	public-unrestricted, public-contextual, community-restricted, initiated, initiated-elder, sacred-secret, numbered 0 to 5	The published scheme: notations 1 to 5 and 99, with no-access as a never-export marker
Namespace	Module prefixes such as `ag:` and `ewe:` with module namespaces	One flat `iroko:` namespace, as published since v1.3.0
Contested knowledge example	`conflictsWith`, `resolutionStatus`, `unresolved`, `assertedBy`	Published terms: `assertionStatus`, `competingClaim`, supporting and rejecting lineages, `authorizedBy`, lineage graphs
Refusal example	`refusalDate`, `refusingAuthority`, `refusalBasis`, `refusalScope`	Published terms: `generatesRefusal`, `involvesAgent`, PROV-O time
Ewé levels	Harvest initiated-only; preparation elder-restricted; activation sacred-secret	Published defaults: harvest and preparation community-only; detailed preparation initiated-only
Governed access	Data described as already "held at irokosociety.org under API-governed access protocols"	Open layer published; governed delivery specified and marked Designed; first build in the Hyatt pilot
Priority claims	"Why Nothing Like This Exists," "no institution of this kind has previously existed," "the theoretical breakthrough," "the first attempt," "the framework is complete"	Restated against named prior work (CARE, Local Contexts, Mukurtu, ODRL, EAD, DCMI) with the framework's particular contribution stated
Standards	CIDOC-CRM listed as an alignment	Not currently referenced in any module; removed from the list
Darwin Core alignment	Plant class declared equivalent to `dwc:Occurrence`	Under review, since an Occurrence is an observation or specimen event rather than a plant or taxon
New sections		VIII Two Governed Corpora; IX Reference Architecture; XII Granularity in v2.0.0; Notes; Appendix D

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